

Assignment 2

1. Show that a function $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous (using any other standard Real Analysis definition) if and only if for every open set $G \subseteq \mathbb{R}$ the preimage $f^{-1}(G)$ is an open set.

Proof. We must first show that $|f(x) - f(y)| < \epsilon$ is equivalent to the condition that $y \in f^{-1}((f(x) - \epsilon, f(x) + \epsilon))$.

($\implies$) Suppose $|f(x) - f(y)| < \epsilon$. Thus:

$$-\epsilon < f(x) - f(y) < \epsilon.$$

$$-\epsilon < f(y) - f(x) < \epsilon \Rightarrow f(x) - \epsilon < f(y) < f(x) + \epsilon.$$

$$f(y) \in (f(x) - \epsilon, f(x) + \epsilon) \Rightarrow y \in f^{-1}((f(x) - \epsilon, f(x) + \epsilon)).$$

($\impliedby$) Suppose $y \in f^{-1}((f(x) - \epsilon, f(x) + \epsilon))$. Thus:

$$f(y) \in (f(x) - \epsilon, f(x) + \epsilon).$$

$$f(x) - \epsilon < f(y) < f(x) + \epsilon \Rightarrow |f(y) - f(x)| < \epsilon.$$

Therefore, $|f(x) - f(y)| < \epsilon$ is equivalent to $y \in f^{-1}((f(x) - \epsilon, f(x) + \epsilon))$.

We know from Real Analysis, that a definition for continuity is as follows:

$$\forall x, y \in \mathbb{R}, \text{ with } \delta, \epsilon > 0, |x - y| < \delta \Rightarrow |f(x) - f(y)| < \epsilon.$$

We must show that these definitions are equivalent.

($\implies$) Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous. Thus:

$$y \in (x - \delta, x + \delta) \Rightarrow y \in f^{-1}((f(x) - \epsilon, f(x) + \epsilon)).$$

Now let $G \subseteq \mathbb{R}$ be open. Therefore G is a union of open intervals, $\cup_{i \in \mathbb{N}} (f(y_i) - \epsilon_i, f(y_i) + \epsilon_i)$. Because f is continuous, we know that in order for $f(x)$ to be within an interval $(f(y_i) - \epsilon_i, f(y_i) + \epsilon_i)$, x must be within some δ_i of y_i . That is, $x \in (y_i - \delta_i, y_i + \delta_i)$. Therefore, given any $f(x)$ in any open interval in G , x must be an element of an open interval in $f^{-1}(G)$. Therefore the preimage of any open interval must be open. Taking a union of open intervals to form any open set, gives the union of open intervals in the domain, which must be open. Therefore f being continuous implies that the preimage of every open set is open.

($\Leftarrow$) Suppose that for every open set $G \subseteq \mathbb{R}$, the preimage $f^{-1}(G)$ is open. Suppose that $f(x) \in G$. Therefore, there exists an open interval $(f(x) - \varepsilon, f(x) + \varepsilon) \subseteq G$, that clearly contains $f(x)$. Let $f(y) \in (f(x) - \varepsilon, f(x) + \varepsilon)$. Because the preimage of this open set is open, it can be represented as an intersection of open intervals. Therefore y must be contained in one of these open intervals. That is, $y \in (f^{-1}(f(x) - \varepsilon, f(x) + \varepsilon))$. Thus, in order for $f(y)$ to be within ε of $f(x)$, it must be the case that y is within some open interval of radius, say δ of x . Therefore, $|f(y) - f(x)| < \varepsilon$ whenever $|y - x| < \delta$. Therefore when the preimage of every open set $G \subseteq \mathbb{R}$ is open, f is continuous.

□

2. Let X be a set and consider the set of subsets of X defined by

$$\mathcal{S} = \{E \subseteq X : E \text{ is countable or } E^C \text{ is countable}\}$$

Prove that $\mathcal{S}$ is a σ -algebra on X .

Proof. We must show that $\mathcal{S}$ is closed under all of the σ -algebra operations.

- (a) $\mathcal{S}$ must contain $\emptyset$. Well, $\emptyset$ is certainly countable, thus $\emptyset \in \mathcal{S}$.
- (b) If E is countable and thus in $\mathcal{S}$, then its complement E^C is also in $\mathcal{S}$, because $(E^C)^C = E \in \mathcal{S}$. Suppose that E^C is countable and thus $E^C \in \mathcal{S}$. Therefore $E \in \mathcal{S}$ because its complement is countable.
- (c) If $X_1, X_2, \dots \in \mathcal{S}$, then we must show that $\bigcup_{i=1}^{\infty} X_i$ must be in $\mathcal{S}$. Suppose that all E_i are uncountable. This means that all E_i^C must be countable otherwise each E_i would not be in $\mathcal{S}$. By DeMorgan's laws we have:

$$\left(\bigcup_{i=1}^{\infty} E_i\right)^C = \bigcap_{i=1}^{\infty} E_i^C$$

Because each E_i^C is countable, the intersection of all $i \in \mathbb{N}$ must also be countable. Thus, $(\bigcup_{i \in \mathbb{N}} E_i)^C$ is countable which implies that its complement, $\bigcup_{i \in \mathbb{N}} E_i \in \mathcal{S}$. Now, suppose that not all E_i are uncountable. The countable intersection described above still contains a countable set, and thus the complement of the infinite union will still be countable, making union be contained in $\mathcal{S}$. Now, suppose that all E_i are countable. Then clearly a countable union of countable sets is countable, so $\bigcup_{i \in \mathbb{N}} E_i \in \mathcal{S}$. We are done.

□

3. Suppose that $\mathcal{S}$ is the smallest σ -algebra on $\mathbb{R}$ containing $\{[r, r+1] : r \in \mathbb{Q}\}$. Prove that $\mathcal{S}$ is equal to the Borel σ -algebra $\mathcal{B}$ on $\mathbb{R}$.

Proof. In order to prove this, we must show that $\mathcal{S} \subseteq \mathcal{B}$ and $\mathcal{B} \subseteq \mathcal{S}$.

($\mathcal{S} \subseteq \mathcal{B}$) We know that $\mathcal{B}$ is the smallest σ -algebra on $\mathbb{R}$ that contains all open sets, and thus all open intervals. Therefore, we must show that we can construct $\{[r, r+1] : r \in \mathbb{Q}\}$ using only the σ -algebra operations on open intervals. Consider the infinite intersection of open intervals:

$$\bigcap_{j=1}^{\infty} (r - \frac{1}{j}, r + 1 + \frac{1}{j}) = [r, r+1], r \in \mathbb{Q}.$$

Therefore $\mathcal{B}$ is a σ -algebra containing $\{[r, r+1] : r \in \mathbb{Q}\}$. Thus $\mathcal{B}$ must contain the smallest σ -algebra containing $\{[r, r+1] : r \in \mathbb{Q}\}$, $\mathcal{S}$. That is, $\mathcal{S} \subseteq \mathcal{B}$.

($\mathcal{B} \subseteq \mathcal{S}$) We know that the Borel σ -algebra, $\mathcal{B}$, is the smallest σ -algebra on $\mathbb{R}$ that contains all open sets, and thus open intervals. We must show that we can construct any open interval (a, b) using the σ -algebra operations on $\{[r, r+1] : r \in \mathbb{Q}\}$. Let $\{a_n\} \in \mathbb{Q}$ be a sequence of rational numbers such that $\{a_n\} \rightarrow a$, and let $\{b_n\} \in \mathbb{Q}$ be a sequence of rational numbers such that $\{b_n\} \rightarrow b$. Note that both sequences are possible due to the density of $\mathbb{Q}$ in $\mathbb{R}$. Furthermore, let $a < \{a_n\} \leq \{b_n\} < b$ and $\{a_n\}, \{b_n\}$ be monotone. We can construct the following:

$$\begin{aligned} \chi &= [(\bigcup_{i=1}^{\infty} [a_i, a_i+1]) \cup (\bigcup_{i=1}^{\infty} [a_i+i, a_i+i+1])] \cap [(\bigcup_{i=1}^{\infty} [b_i-i-1, b_i-1]) \cup (\bigcup_{i=1}^{\infty} [b_i-1, b_i])]. \\ &\implies \chi = (a, \infty) \cap (-\infty, b) = (a, b). \end{aligned}$$

Thus, we have constructed an arbitrary open interval from $\{[r, r+1] : r \in \mathbb{Q}\}$. Because $\mathcal{B}$ is the smallest σ -algebra containing all open intervals, $\mathcal{S}$ must contain $\mathcal{B}$. Therefore $\mathcal{B} \subseteq \mathcal{S}$.

Because $\mathcal{B} \subseteq \mathcal{S}$ and $\mathcal{S} \subseteq \mathcal{B}$, $\mathcal{S} = \mathcal{B}$. □

4. Give an example of a set that is Borel but is neither open or closed.

$\mathbb{Q}$ is countable, and thus Borel, however it is not closed because it does not contain all of its limit points, any irrational number for example, and its complement is not closed, implying it cannot be open.